

Probability

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Lecture 1 – 23/01/26

§1 Fundamentals

Definition 1.1. Let Ω be a set and $\mathcal{F}$ is a collection of subsets of Ω . We call $\mathcal{F}$ a σ -algebra if

- $\Omega \in \mathcal{F}$;
- if $A \in \mathcal{F}$, then $A^c \in \mathcal{F}$;
- if $(A_n)_{n \in \mathbb{N}}$ is a countable collection of sets in $\mathcal{F}$, then

$$\bigcup A_n \in \mathcal{F}$$

Definition 1.2. Let $\mathcal{F}$ be a σ -algebra on Ω . A function

$$\mathbb{P} : \mathcal{F} \rightarrow [0, 1]$$

is called a *probability measure* if

- $\mathbb{P}(\Omega) = 1$;

- for any countable disjoint collection $(A_n)_{n \in \mathbb{N}}$ in $\mathcal{F}$,

$$\mathbb{P}\left(\bigcup A_n\right) = \sum \mathbb{P}(A_n)$$

We call $\mathbb{P}(A)$ the *probability* of A .

Then $(\Omega, \mathcal{F}, \mathbb{P})$ is a *probability space*.

Definition 1.3. The elements of Ω are called *outcomes*, and the elements of the σ -algebra $\mathcal{F}$ are called *events*.

Importantly, we talk about probabilities of *events*, never probabilities of outcomes – the domain of $\mathbb{P}$ is $\mathcal{F}$, not Ω .

Remark. When Ω is countable, we take $\mathcal{F}$ to be all subsets of Ω . However, this framework will become useful when considering uncountable Ω .

Let us note some important properties of probability measures that follow from the definition.

- $\mathbb{P}(A^c) = 1 - \mathbb{P}(A)$;
- $\mathbb{P}(\emptyset) = 0$;
- $\mathbb{P}(A) \leq \mathbb{P}(B)$ if $A \subseteq B$;
- $\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B)$.

Example 1.4 • *Rolling a fair die.* Then $\Omega = \{1, 2, 3, 4, 5, 6\}$, $\mathcal{F} = \mathcal{P}(\Omega)$, and

$$\mathbb{P}(A) = \frac{|A|}{6}$$

since all outcomes are equally likely.

- $\Omega = \{\omega_1, \dots, \omega_n\}$, $\mathcal{F} = \mathcal{P}(\Omega)$, $\mathbb{P}(A) = \frac{|A|}{|\Omega|}$ models the experiment of picking an element of Ω uniformly at random.
- *Picking balls from a bag.* Suppose we have n balls labelled $\{1, \dots, n\}$, and we pick $k \leq n$ balls uniformly at random without replacement. Then $\Omega = \{A : A \subseteq \{1, \dots, n\}, |A| = k\}$, so $|\Omega| = \binom{n}{k}$ and $\mathbb{P}(\{\omega\}) = \binom{n}{k}^{-1}$ for any $\omega \in \Omega$.

- *A deck of 52 cards.* Take a well-shuffled deck of 52 cards. Then $\Omega = \{\text{all permutations of 52 cards}\}$, so $|\Omega| = 52!$.

What is, for example, $\mathbb{P}(\text{top 2 cards are aces})$? It is $\frac{4 \cdot 3 \cdot 50!}{52!}$.

- *Largest digit.* Consider a string of n random digits from 0 up to 9. Then $\Omega = \{0, \dots, 9\}^n$, $|\Omega| = 10^n$.

Consider the events $A_k = \{\text{no digit exceeds } k\}$, $B_k = \{\text{largest digit is } k\}$. Then

$$\begin{aligned} \mathbb{P}(A_k) &= \frac{(k+1)^n}{10^n} \\ \implies \mathbb{P}(B_k) &= \mathbb{P}(A_k \setminus A_{k-1}) = \frac{(k+1)^n - k^n}{10^n} \end{aligned}$$

- *Birthday problem.* There are n people. What is the probability that at least 2 of them share the same birthday?

Assume each birthday is equally likely to be one of $\{1, \dots, 365\}$. Then $\Omega = \{1, \dots, 365\}^n$, $|\Omega| = 365^n$, and $\mathbb{P}(\{\omega\}) = \frac{1}{365^n}$.

Let $A = \{\text{at least 2 people have the same birthday}\}$. Instead we count $A^c = \{\text{all birthdays different}\}$.

$$\begin{aligned} \mathbb{P}(A^c) &= \frac{365 \cdot 364 \cdots (365 - n + 1)}{365^n} \\ \implies \mathbb{P}(A) &= 1 - \frac{365 \cdot 364 \cdots (365 - n + 1)}{365^n} \end{aligned}$$

Perhaps surprisingly, if $n = 22$, $\mathbb{P}(A) = 0.476$, while if $n = 23$, $\mathbb{P}(A) = 0.507$.

§1.1 Combinatorial analysis

Let us introduce some notions.

1. Let Ω be a finite set with $|\Omega| = n$. We want to partition Ω into k disjoint subsets $\Omega_1, \dots, \Omega_k$ with $|\Omega_i| = n_i$ and $\sum n_i = n$. How many ways are there to do this? It is

$$\begin{aligned}
 M &= \binom{n}{n_1} \cdot \binom{n-n_1}{n_2} \cdots \binom{n-(n_1+\cdots+n_{k-1})}{n_k} \\
 &= \frac{n!}{n_1!n_2!\cdots n_k!} \\
 &= \binom{n}{n_1, n_2, \dots, n_k}
 \end{aligned}$$

This is the *multinomial coefficient*.

2. Suppose $f : \{1, \dots, k\} \rightarrow \{1, \dots, n\}$ (with $k \leq n$). f is *strictly increasing* if $x < y \implies f(x) < f(y)$, and f is *increasing* if $x < y \implies f(x) \leq f(y)$.

How many strictly increasing $f : \{1, \dots, k\} \rightarrow \{1, \dots, n\}$ are there? Such a function is determined uniquely by its range, so there are $\binom{n}{k}$ of them.

What about increasing functions? Let's define a bijection

$$\{f : \{1, \dots, k\} \rightarrow \{1, \dots, n\} \text{ inc.}\} \rightarrow \{f : \{1, \dots, k\} \rightarrow \{1, n+k-1\} \text{ strictly inc.}\}$$

which takes f to g with $g(i) = f(i) + i - 1$. Hence we have $\binom{n+k-1}{k}$ such functions.

Definition 1.5 (Asymptotic notation). If (a_n) and (b_n) are sequences, we write $a_n \sim b_n$ if

$$\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = 1$$

§1.2 Stirling's formula

Theorem 1.6 (Stirling)

$$n! \sim n^n \sqrt{2\pi n} e^{-n}$$

as $n \rightarrow \infty$.

Weaker statement. We shall first show a weaker statement, that $\log n! \sim n \log n$.

Define $l_n = \log n! = \log 2 + \cdots + \log n$. We have

$$\log \lfloor x \rfloor \leq \log x \leq \log \lfloor x+1 \rfloor$$

Integrating from 1 to n gives

$$\begin{aligned}
 \sum_{k=1}^{n-1} \log k &\leq \int_1^n \log x \, dx \leq \sum_{k=1}^n \log k \\
 \implies l_{n-1} &\leq n \log n - n + 1 \leq l_n \\
 \implies n \log n - n + 1 &\leq l_n \leq (n+1) \log(n+1) - (n+1) + 1
 \end{aligned}$$

Now the result follows by dividing through by $n \log n$ and taking the limit. ■

Non-examinable proof of Stirling. For every twice-differentiable $f : \mathbb{R} \rightarrow \mathbb{R}$, for all $a < b$, we have

$$\int_a^b f(x) dx = \frac{f(a) + f(b)}{2}(b - a) - \frac{1}{2} \int_a^b (x - a)(b - x) f''(x) dx$$

obtained by integrating twice by parts. Put $f = \log$, $a = k$, $b = k + 1$ for $k \in \mathbb{N}$, to get

$$\int_k^{k+1} \log x dx = \frac{\log k + \log(k + 1)}{2} + \frac{1}{2} \int_k^{k+1} \frac{(x - k)(k + 1 - x)}{x^2} dx$$

Summing from $k = 1$ to $n - 1$, we have

$$\int_1^n \log x dx = \frac{\log(n - 1)! + \log n!}{2} + \sum_{k=1}^{n-1} a_k$$

where $a_k = \frac{1}{2} \int_0^1 \frac{x(1-x)}{(x+k)^2} dx$. Hence

$$n \log n - n + 1 = \log n! - \frac{\log n}{2} + \sum_{k=1}^{n-1} a_k$$

and so

$$\begin{aligned} \log n! &= n \log n - n + 1 + \frac{\log n}{2} - \sum_{k=1}^{n-1} a_k \\ \implies n! &= n^n \cdot e^{-n} \cdot \sqrt{n} \cdot \exp \left(1 - \sum_{k=1}^{n-1} a_k \right) \end{aligned}$$

Let us now consider

$$\begin{aligned} a_k &= \frac{1}{2} \int_0^1 \frac{x(1-x)}{(x+k)^2} dx \\ &\leq \frac{1}{2} \int_0^1 \frac{1}{2k^2} \int_0^1 x(1-x) dx \\ &= \frac{1}{12k^2} \end{aligned}$$

and so $\sum_{k=1}^{\infty} a_k < \infty$. Now put

$$\begin{aligned} A &= \exp \left(1 - \sum_{k=1}^{\infty} a_k \right) \\ \implies n! &= n^n \cdot e^{-n} \cdot \sqrt{n} \cdot A \cdot \underbrace{\exp \left(\sum_{k=n}^{\infty} a_k \right)}_{\rightarrow 1 \text{ as } n \rightarrow \infty} \\ \implies \frac{n!}{n^n e^{-n} \sqrt{n}} &\rightarrow A \text{ as } n \rightarrow \infty \end{aligned}$$

It remains to show that $A = \sqrt{2\pi}$. Consider

$$2^{-2n} \binom{2n}{n} = 2^{-2n} \frac{(2n)!}{(n!)^2}$$

$$\begin{aligned} &\sim \frac{2^{-2n}(2n)^{2n}e^{-2n}\sqrt{2n}A}{n^{2n}e^{-2n}nA^2} \\ &= \frac{\sqrt{2}}{A\sqrt{n}} \end{aligned}$$

Now, we shall use a different method to show

$$2^{-2n} \binom{2n}{n} \sim \frac{1}{\sqrt{\pi n}} \text{ as } n \rightarrow \infty$$

Consider

$$I_n = \int_0^{\pi/2} \cos^n \theta \, d\theta$$

We have $I_0 = \frac{\pi}{2}$, $I_1 = 1$, and, by integrating by parts, we have

$$I_n = \frac{n-1}{n} I_{n-2} \quad (*)$$

so that

$$\begin{aligned} I_{2n} &= \frac{2n-1}{2n} I_{2n-2} \\ &= \dots = \frac{(2n-1)(2n-3)\dots 3 \cdot 1}{(2n)(2n-2)\dots 2} \cdot \frac{\pi}{2} \\ &= \frac{2n(2n-1)(2n-2)\dots 3 \cdot 2 \cdot 1}{2n \cdot 2n \cdot (2n-2) \cdot (2n-2) \dots 2 \cdot 2} \cdot \frac{\pi}{2} \\ &= \frac{(2n)!}{2^{2n}(n!)^2} \cdot \frac{\pi}{2} \\ &= \frac{\pi}{2} \cdot 2^{-2n} \binom{2n}{n} \end{aligned}$$

and we can similarly obtain

$$\begin{aligned} I_{2n+1} &= \frac{2n \dots 4 \cdot 2}{(2n+1) \dots 3 \cdot 1} \cdot 1 \\ &= \frac{1}{2n+1} \left(2^{-2n} \binom{2n}{n} \right)^{-1} \end{aligned}$$

Now, we want to show that $\frac{I_{2n}}{I_{2n+1}} \rightarrow 1$ as $n \rightarrow \infty$. By (*), we have

$$\frac{I_n}{I_{n-2}} \rightarrow 1 \text{ as } n \rightarrow \infty$$

But $I_n = \int_0^{\pi/2} \cos^n \theta \, d\theta$, so I_n is decreasing in n . This gives

$$\frac{I_{2n}}{I_{2n-2}} \leq \frac{I_{2n}}{I_{2n+1}} \leq \frac{I_{2n-1}}{I_{2n+1}}$$

which shows the result by sandwich. Hence, we get, as $n \rightarrow \infty$,

$$\begin{aligned} &\frac{\frac{\pi}{2} \cdot 2^{-2n} \cdot \binom{2n}{n}}{\frac{1}{2n+1} \cdot \left(2^{-2n} \binom{2n}{n} \right)^{-1}} \rightarrow 1 \\ \implies &\left(2^{-2n} \binom{2n}{n} \right)^2 \sim \frac{1}{\pi n} \end{aligned}$$

■

Proposition 1.7 (Countable subadditivity)

Let (A_n) be a collection of events in $\mathcal{F}$. Then

$$\mathbb{P}\left(\bigcup A_n\right) \leq \sum \mathbb{P}(A_n)$$

Proof. Let $B_1 = A_1$, and for $n \geq 2$, let $B_n = A_n \setminus (A_1 \cup \cdots \cup A_{n-1})$. By definition, it's immediate that (B_n) is a disjoint collection of sets in $\mathcal{F}$, and $\bigcup B_n = \bigcup A_n$, so

$$\begin{aligned} \mathbb{P}\left(\bigcup A_n\right) &= \mathbb{P}\left(\bigcup B_n\right) = \sum \mathbb{P}(B_n) \\ &\leq \sum \mathbb{P}(A_n) \end{aligned}$$

since $B_n \subseteq A_n$ for all n . ■

Proposition 1.8 (Continuity of probability measures)

Take a sequence of events $(A_n)_{n \in \mathbb{N}}$ such that $A_n \subseteq A_{n+1}$ for all n – we say (A_n) is an *increasing* sequence.

Then $\mathbb{P}(A_n)$ is increasing and converges. In particular,

$$\mathbb{P}(A_n) \rightarrow \mathbb{P}\left(\bigcup A_n\right)$$

as $n \rightarrow \infty$.

Proof. Define $B_1 = A_1$, and for $n \geq 2$, let $B_n = A_n \setminus (A_1 \cup \cdots \cup A_{n-1})$. Then

$$\bigcup_{k=1}^n B_k = A_n$$

and the B_n 's are disjoint. Then

$$\begin{aligned} \mathbb{P}(A_n) &= \mathbb{P}\left(\bigcup_{k=1}^n B_k\right) \\ &= \sum_{k=1}^n \mathbb{P}(B_k) \rightarrow \sum_{k=1}^{\infty} \mathbb{P}(B_k) \end{aligned}$$

Also,

$$\begin{aligned} \sum_{k=1}^{\infty} \mathbb{P}(B_k) &= \mathbb{P}\left(\bigcup B_k\right) \\ &= \mathbb{P}(A_k) \end{aligned}$$

■

Corollary 1.9

Analogously, take a sequence of events (A_n) such that $A_{n+1} \subseteq A_n$ for all n – we say (A_n) is a decreasing sequence. Then

$$\mathbb{P}(A_n) \rightarrow \mathbb{P}\left(\bigcap A_n\right)$$

Proof. Take complements and use the previous claim. ■

Proposition 1.10 (Inclusion-exclusion formula for probability measures)

Let $A_1, \dots, A_n \in \mathcal{F}$. Then

$$\mathbb{P}\left(\bigcup A_i\right) = \sum_{k=1}^n (-1)^{k+1} \sum_{1 \leq i_1 < \dots < i_k \leq n} \mathbb{P}(A_{i_1} \cap \dots \cap A_{i_k})$$

Proof. We proceed by induction.

Base case. ADD

Inductive step. We have

$$\begin{aligned} \mathbb{P}\left(\bigcup_{i=1}^n A_i\right) &= \mathbb{P}\left(\left(\bigcup_{i=1}^{n-1} A_i\right) \cup A_n\right) \\ &= \mathbb{P}\left(\bigcup_{i=1}^{n-1} A_i\right) + \mathbb{P}(A_n) - \mathbb{P}\left(\bigcup_{i=1}^{n-1} A_i \cap A_n\right) \\ &= \mathbb{P}\left(\bigcup_{i=1}^{n-1} A_i\right) + \mathbb{P}(A_n) - \mathbb{P}\left(\left(\bigcup_{i=1}^{n-1} A_i\right) \cap A_n\right) \\ &= \mathbb{P}\left(\bigcup_{i=1}^{n-1} A_i\right) + \mathbb{P}(A_n) - \mathbb{P}\left(\bigcup_{i=1}^{n-1} \underbrace{(A_i \cap A_n)}_{B_i}\right) \end{aligned}$$

Applying the inductive hypothesis, we get

$$\begin{aligned} \mathbb{P}\left(\bigcup_{i=1}^{n-1} A_i\right) &= \sum_{k=1}^{n-1} (-1)^{k+1} \sum_{1 \leq i_1 < \dots < i_k \leq n-1} \mathbb{P}(A_{i_1} \cap \dots \cap A_{i_k}) \\ \mathbb{P}\left(\bigcup_{i=1}^{n-1} B_i\right) &= \sum_{k=1}^{n-1} (-1)^{k+1} \sum_{1 \leq i_1 < \dots < i_k \leq n-1} \mathbb{P}(A_{i_1} \cap \dots \cap A_{i_k} \cap A_n) \end{aligned}$$

and subbing this in, we will find that everything works out. ■

Proposition 1.11 (Bonferroni inequalities)

Let $A_1, \dots, A_n$ be events. Then

$$\mathbb{P}\left(\bigcup A_i\right) \begin{cases} \leq \sum_{k=1}^r (-1)^{k+1} \sum_{1 \leq i_1 < \dots < i_k \leq n} \mathbb{P}(A_{i_1} \cap \dots \cap A_{i_k}) & \text{if } r \text{ odd} \\ \geq \sum_{k=1}^r (-1)^{k+1} \sum_{1 \leq i_1 < \dots < i_k \leq n} \mathbb{P}(A_{i_1} \cap \dots \cap A_{i_k}) & \text{if } r \text{ even} \end{cases}$$

where those sums are just truncated expansions of inclusion-exclusion.

Proof. We induct again.

Base case. $\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B) \leq \mathbb{P}(A) + \mathbb{P}(B)$

Inductive step. Suppose r is odd. Then

$$\mathbb{P}\left(\bigcup_{i=1}^n A_i\right) = \mathbb{P}\left(\bigcup_{i=1}^{n-1} A_i\right) + \mathbb{P}(A_n) - \mathbb{P}\left(\bigcup_{i=1}^{n-1} \underbrace{(A_i \cap A_n)}_{B_i}\right)$$

By the inductive hypotheses,

$$\begin{aligned} \mathbb{P}\left(\bigcup_{i=1}^{n-1} A_i\right) &\leq \sum_{k=1}^r (-1)^{k+1} \sum_{1 \leq i_1 < \dots < i_k \leq n-1} \mathbb{P}(A_{i_1} \cap \dots \cap A_{i_k}) \\ \mathbb{P}\left(\bigcup_{i=1}^{n-1} (A_i \cap A_n)\right) &\geq \sum_{k=1}^{r-1} (-1)^{k+1} \sum_{1 \leq i_1 < \dots < i_k \leq n-1} \mathbb{P}(A_{i_1} \cap \dots \cap A_{i_k} \cap A_n) \end{aligned}$$

where in the former, we choose to apply the identity for odd r , and in the latter, we apply it for even $r-1$. Substituting much as for the proof of the inclusion-exclusion formula, we obtain the result. The even case is analogous. ■

Let us now consider the special case where Ω is a finite set and $\mathbb{P}(A) = |A| / |\Omega|$. Then we have

Corollary 1.12 (Inclusion-exclusion principle for sets)

Suppose Ω is finite. Then

$$|A_1 \cup \dots \cup A_n| = \sum_{k=1}^n (-1)^{k+1} \sum_{1 \leq i_1 < \dots < i_k \leq n} |A_{i_1} \cap \dots \cap A_{i_k}|$$

Proof. Consider the probability space $(\Omega, \mathcal{F}, \mathbb{P})$ where $\mathbb{P}(A) = |A| / |\Omega|$. Then, apply inclusion-exclusion and cancel $1/|\Omega|$ from both sides. ■

§2 Basic applications

§2.1 Counting surjections

Define a probability space as follows:

$$\Omega = \{f : \{1, \dots, n\} \rightarrow \{1, \dots, m\}\}$$

$$A = \{f \in \Omega : f \text{ is a surjection}\}$$

What is $|A|$? Let us define a set of events

$$A_i = \{f \in \Omega : i \notin \{f(1), \dots, f(n)\}\}, i \in \{1, \dots, m\}$$

so that

$$\begin{aligned} A &= A_1^c \cap \dots \cap A_m^c = (A_1 \cup \dots \cup A_m)^c \\ \Rightarrow |A| &= |\Omega| - |A_1 \cup \dots \cup A_m| \\ &= |\Omega| - \sum_{k=1}^m (-1)^{k+1} \sum_{1 \leq i_1 < \dots < i_k \leq m} |A_{i_1} \cap \dots \cap A_{i_k}| \\ &= \underbrace{|\Omega|}_{m^n} - \sum_{k=1}^m (-1)^{k+1} \binom{m}{k} (m-k)^n \\ &= \sum_{k=0}^m (-1)^k \binom{m}{k} (m-k)^n \end{aligned}$$

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§2.2 Counting derangements

Definition 2.1. A *derangement* is a permutation with no fixed points.

Let

$$\begin{aligned} \Omega &= \{\text{permutations of } \{1, \dots, n\}\} \\ A &= \{\text{derangements}\} \end{aligned}$$

Let

$$A_i = \{f \in \Omega : f(i) = i\}$$

so that

$$\begin{aligned} A &= A_1^c \cap \dots \cap A_n^c = (A_1 \cup \dots \cup A_n)^c \\ \Rightarrow \mathbb{P}(A) &= 1 - \mathbb{P}\left(\bigcup A_i\right) = 1 - \sum_{k=1}^n (-1)^{k+1} \sum_{1 \leq i_1 < \dots < i_k \leq n} \underbrace{\mathbb{P}(A_{i_1} \cap \dots \cap A_{i_k})}_{\frac{(n-k)!}{n!}} \\ &= 1 - \sum_{k=1}^n (-1)^{k+1} \binom{n}{k} \frac{(n-k)!}{n!} \\ &= 1 - \sum_{k=1}^n (-1)^{k+1} \frac{1}{k!} \\ &= \sum_{k=0}^n \frac{(-1)^k}{k!} \rightarrow e^{-1} \end{aligned}$$

§3 Independence and conditional probability

Definition 3.1 (Independence). Let $A, B \in \mathcal{F}$. We say A and B are *independent* and write $A \perp B$ if

$$\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B)$$

Likewise, a countable collection of events $(A_n)_{n \in \mathbb{N}}$ is said to be independent if, for distinct $i_1, \dots, i_k$,

$$\mathbb{P}(A_{i_1} \cap \dots \cap A_{i_k}) = \prod_{j=1}^k \mathbb{P}(A_{i_j})$$

Remark. Pairwise independence of events does not imply independence.

Example

Toss a fair coin twice. We have

$$\begin{aligned} \Omega &= \{(0, 0), (0, 1), (1, 0), (1, 1)\} \\ \mathbb{P}(\{\omega\}) &= \frac{1}{4} \end{aligned}$$

Let

$$\begin{aligned} A &= \{(0, 0), (0, 1)\} \\ B &= \{(0, 0), (1, 0)\} \\ C &= \{(1, 0), (0, 1)\} \end{aligned}$$

We have $\mathbb{P}(A) = \mathbb{P}(B) = \mathbb{P}(C)$, and furthermore we can check that $A \perp B$, $A \perp C$ and $B \perp C$. But $\mathbb{P}(A \cap B \cap C) = \mathbb{P}(\emptyset) = 0$, so they are not independent.

Proposition 3.2

If A is independent of B , then A is independent of B^c .

Proof. We have

$$\begin{aligned} \mathbb{P}(A \cap B^c) &= \mathbb{P}(A) - \mathbb{P}(A \cap B) \\ &= \mathbb{P}(A)(1 - \mathbb{P}(B)) \\ &= \mathbb{P}(A)\mathbb{P}(B^c) \end{aligned}$$

■

Definition 3.3 (Conditional probability). Take a probability space $(\Omega, \mathcal{F}, \mathbb{P})$ and $B \in \mathcal{F}$ with $\mathbb{P}(B) > 0$. For $A \in \mathcal{F}$ we write the *conditional probability of A given B*

$$\mathbb{P}(A \mid B) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)}$$

Remark. Notably, if $A \perp B$, then $\mathbb{P}(A | B) = \mathbb{P}(A)$, as we would expect.

Proposition 3.4 (Countable additivity for conditional probability)

Suppose $(A_n)_{n \in \mathbb{N}}$ is a disjoint sequence in $\mathcal{F}$. Then

$$\mathbb{P}\left(\bigcup A_n \mid B\right) = \sum \mathbb{P}(A_n \mid B)$$

Proof. We have

$$\begin{aligned} \mathbb{P}\left(\bigcup A_n \mid B\right) &= \frac{\mathbb{P}(\bigcup A_n \cap B)}{\mathbb{P}(B)} \\ &= \frac{\mathbb{P}(\bigcup (A_n \cap B))}{\mathbb{P}(B)} \\ &= \sum \frac{\mathbb{P}(A_n \cap B)}{\mathbb{P}(B)} \\ &= \sum \mathbb{P}(A_n \mid B) \end{aligned}$$

■

Proposition 3.5 (Law of total probability)

Suppose $(B_n)_{n \in \mathbb{N}}$ is a disjoint collection in $\mathcal{F}$, $\bigcup B_n = \Omega$, and $\mathbb{P}(B_n) > 0$ for every n . Let $A \in \mathcal{F}$, then

$$\mathbb{P}(A) = \sum_n \mathbb{P}(A \mid B_n) \mathbb{P}(B_n)$$

Proof. We have

$$\begin{aligned} \mathbb{P}(A) &= \mathbb{P}\left(A \cap \left(\bigcup B_n\right)\right) \\ &= \mathbb{P}\left(\bigcup (B_n \cap A)\right) \\ &= \sum \mathbb{P}(B_n \cap A) \\ &= \sum \mathbb{P}(A \mid B_n) \mathbb{P}(B_n) \end{aligned}$$

■

Proposition 3.6 (Bayes' theorem)

Suppose (B_n) are disjoint, $\bigcup B_n = \Omega$ and $\mathbb{P}(B_n) > 0$ for every n . Let $A \in \mathcal{F}$, then

$$\mathbb{P}(B_n \mid A) = \frac{\mathbb{P}(A \mid B_n) \cdot \mathbb{P}(B_n)}{\sum_k \mathbb{P}(A \mid B_k) \mathbb{P}(B_k)}$$

Proof sketch. Use the definitions of $\mathbb{P}(B_n | A)$ and $\mathbb{P}(A | B_n)$, and expand the bottom using the law of total probability. ■

This formula is the basis of so-called Bayesian statistics. We know the probability of $\mathbb{P}(B_n)$ and we have a model which gives us $\mathbb{P}(A | B_n)$. With the formula, we find the *posterior* probability of B_n given that A occurred.

Example (False positives for a rare condition)

Suppose that condition A affects 0.1% of the population. We have a medical test which is positive for 98% of the affected population and 1% of the unaffected population. What is the probability that a random individual who tests positive has the disease?

Put $A = \{\text{individual suffers from } A\}$, $P = \{\text{individual tests positive}\}$. We have $\mathbb{P}(A) = 0.001$, $\mathbb{P}(P | A) = 0.98$, $\mathbb{P}(P | A^c) = 0.01$, so

$$\begin{aligned}\mathbb{P}(A | P) &= \frac{\mathbb{P}(P | A)\mathbb{P}(A)}{\mathbb{P}(P | A)\mathbb{P}(A) + \mathbb{P}(P | A^c)\mathbb{P}(A^c)} \\ &= 0.089\end{aligned}$$

In particular, we can see

$$\mathbb{P}(A | P) = \frac{1}{1 + \frac{\mathbb{P}(P|A^c)\mathbb{P}(A^c)}{\mathbb{P}(P|A)\mathbb{P}(A)}}$$

Since $\mathbb{P}(A^c)$ and $\mathbb{P}(P | A)$ are both very close to one, the relevant ratio is

$$\frac{\mathbb{P}(P | A^c)}{\mathbb{P}(A)}$$

But since $\mathbb{P}(P | A^c) \gg \mathbb{P}(A)$, this causes a very small value for $\mathbb{P}(A | P)$.

Example • What is the probability that I have two boys given that I have two children one of whom is a boy?

Since no further information is given, we take all outcomes to be equally likely. Let

$$BG = \{\text{elder child a boy, younger a girl}\}$$

with GB, BB, GG defined analogously. Then we want

$$\begin{aligned} \mathbb{P}(BB \mid BB \cup BG \cup GB) &= \frac{\mathbb{P}(BB)}{\mathbb{P}(BB \cup BG \cup GB)} \\ &= \frac{\frac{1}{4}}{\frac{3}{4}} = \frac{1}{3} \end{aligned}$$

- What about $\mathbb{P}(\text{both boys} \mid \text{eldest a boy})$? This is

$$\mathbb{P}(BB \mid BB \cup BG) = \frac{1}{2}$$

- What about $\mathbb{P}(\text{both boys} \mid \text{one a boy born on a Thursday})$? Denote a boy born on Thursday by T, a boy not born on Thursday by N, and a girl by G. Then we want

$$\begin{aligned} &\mathbb{P}(TT \cup TN \cup NT \mid TT \cup TG \cup GT \cup TN \cup NT) \\ &= \frac{\mathbb{P}(TT \cup TN \cup NT)}{\mathbb{P}(TT \cup TG \cup GT \cup TN \cup NT)} \\ &= \frac{\left(\frac{1}{2} \cdot \frac{1}{7}\right)^2 + \left(\frac{1}{2} \cdot \frac{1}{7}\right) \left(\frac{1}{2} \cdot \frac{6}{7}\right) + \left(\frac{1}{2} \cdot \frac{6}{7}\right) \left(\frac{1}{2} \cdot \frac{1}{7}\right)}{\left(\frac{1}{2} \cdot \frac{1}{7}\right)^2 + \left(\frac{1}{2} \cdot \frac{1}{7}\right) \cdot \frac{1}{2} + \frac{1}{2} \cdot \left(\frac{1}{2} \cdot \frac{1}{7}\right) + \left(\frac{1}{2} \cdot \frac{1}{7}\right) \left(\frac{1}{2} \cdot \frac{6}{7}\right) + \left(\frac{1}{2} \cdot \frac{6}{7}\right) \left(\frac{1}{2} \cdot \frac{1}{7}\right)} \\ &=; \frac{13}{27} \end{aligned}$$

Morally speaking, we're providing information that helps specify the boy in question more, which brings the probability closer to a half.

Example (Simpson's paradox)

Consider the following data:

<i>All applicants</i>	Admitted	Rejected	% Admitted
State	25	25	50%
Indep.	28	22	56%

<i>London schools</i>	Admitted	Rejected	% Admitted
State	15	22	41%
Indep.	5	8	38%

<i>Cambridge schools</i>	Admitted	Rejected	% Admitted
State	10	3	77%
Indep.	23	14	62%

– paradoxically, the state admission rate is higher when the school locations are taken individually, but the independent admission right is higher for the combined data. This phenomenon is called *confounding*, and arises when we aggregate data from disparate populations.

Let

$$\begin{aligned} A &= \{\text{individual is admitted}\} \\ B &= \{\text{individual from London}\} \\ C &= \{\text{individual from state school}\} \end{aligned}$$

Then

$$\begin{aligned} \mathbb{P}(A \mid B \cap C) &> \mathbb{P}(A \mid B \cap C^c) \\ \mathbb{P}(A \mid B^c \cap C) &> \mathbb{P}(A \mid B^c \cap C^c) \\ \text{but } \mathbb{P}(A \mid C) &< \mathbb{P}(A \mid C^c) \end{aligned}$$

Let's examine how this can happen. We have

$$\begin{aligned} \mathbb{P}(A \mid C) &= \mathbb{P}(A \cap B \mid C) + \mathbb{P}(A \cap B^c \mid C) \\ &= \frac{\mathbb{P}(A \cap B \cap C)}{\mathbb{P}(C)} + \frac{\mathbb{P}(A \cap B^c \cap C)}{\mathbb{P}(C)} \\ &= \frac{\mathbb{P}(A \mid B \cap C)\mathbb{P}(B \cap C)}{\mathbb{P}(C)} + \frac{\mathbb{P}(A \mid B^c \cap C)\mathbb{P}(B^c \cap C)}{\mathbb{P}(C)} \\ &= \mathbb{P}(A \mid B \cap C)\mathbb{P}(B \mid C) + \mathbb{P}(A \mid B^c \cap C)\mathbb{P}(B^c \mid C) \\ &> \mathbb{P}(A \mid B \cap C^c)\mathbb{P}(B \mid C) + \mathbb{P}(A \mid B^c \cap C^c)\mathbb{P}(B^c \mid C) \end{aligned}$$

by the first two inequalities from above. Now, if $\mathbb{P}(B \mid C) = \mathbb{P}(B \mid C^c)$, then we would get $\mathbb{P}(A \mid C) > \mathbb{P}(A \mid C^c)$, which is the intuitive expectation – but here this does not hold, and so we can make the ‘paradox’ hold.

Even more simply, this is just the fact that, if $\frac{a_1}{a_2} > \frac{b_1}{b_2}$ and $\frac{c_1}{c_2} > \frac{d_1}{d_2}$, it does not necessarily hold that $\frac{a_1+c_1}{a_2+c_2} > \frac{b_1+d_1}{b_2+d_2}$.

§4 Probability distributions

§4.1 Discrete probability distributions

Take a probability space $(\Omega, \mathcal{F}, \mathbb{P})$ with Ω either finite or countable – that is, $\Omega = \{\omega_1, \omega_2, \dots\}$. As normal, we take $\mathcal{F} = \mathcal{P}(\Omega)$, and so we need only care about $\mathbb{P}(\{\omega_i\})$ for every i .

Definition 4.1 (Discrete probability distributions). For such a setup, we call the sequence of probabilities $(\mathbb{P}(\{\omega_i\}))_i$ a *discrete probability distribution*. We may also write $p_i = \mathbb{P}(\{\omega_i\})$.

Let us consider some examples of discrete probability distributions.

- ① *Bernoulli distribution*. The Bernoulli distribution models the outcome of a toss of a biased coin. We have $\Omega = \{0, 1\}$, $p_0 = \mathbb{P}(\{0\}) = 1 - p$, $p_1 = \mathbb{P}(\{1\}) = p$ for some p with $p \in [0, 1]$.

- ② *Binomial distribution*. We denote this $\text{Bin}(n, p)$, with $n \in \mathbb{N}$, $p \in [0, 1]$. It models the number of heads seen for n independent tosses of a p -biased coin. We have

$$\Omega = \{0, 1, \dots, n\}$$

$$p_k = \binom{n}{k} p^k (1 - p)^{n-k}$$

- ③ *Multinomial distribution*. We denote this $\text{M}(n, p_1, \dots, p_k)$, where $p_i \geq 0$, $\sum p_i = 1$. Given k boxes and n balls placed in box i with p_i , the multinomial distribution gives the probability of n_l balls in box l for every n . We have

$$\Omega = \left\{ (n_1, \dots, n_k) \in \mathbb{N}^k : \sum_{i=1}^k n_i = n \right\}$$

$$\mathbb{P}(n_1, \dots, n_k) = \binom{n}{n_1} p_1^{n_1} \cdot \binom{n - n_1}{n_2} p_2^{n_2} \cdots \binom{n - n_1 - \dots - n_{k-1}}{n_k} p_k^{n_k}$$

$$= \binom{n}{n_1, n_2, \dots, n_k} p_1^{n_1} p_2^{n_2} \cdots p_k^{n_k}$$

- ④ *Geometric distribution*. The geometric distribution gives the probability that the first heads achieved on a p -coin is after k tosses:

$$\Omega = \{1, 2, \dots\}$$

$$p_k = (1 - p)^{k-1} p$$

Alternatively, we might want to model the probability that we see k tails before the first heads:

$$\Omega = \{0, 1, \dots\}$$

$$p'_k = (1 - p)^k p$$

- ⑤ *Poisson distribution*. Suppose customers arrive into a shop in $[0, 1]$. Discretise $[0, 1]$ into $\left\{ \left[\frac{i-1}{N}, \frac{i}{N} \right] \right\}$, $i = 1, \dots, N$, $N \in \mathbb{N}$. Independently, in each interval, a customer arrives with probability p , and doesn't arrive with probability $1 - p$.

Then

$$\mathbb{P}(k \text{ customers have arrived}) = \binom{N}{k} p^k (1-p)^{N-k}$$

– in other words, this follows a binomial distribution. Now, take $p = \frac{\lambda}{N}$, so

$$\begin{aligned} \mathbb{P}(k \text{ customers have arrived}) &= \frac{N!}{k!(N-k)!} \left(\frac{\lambda}{N}\right)^k \left(1 - \frac{\lambda}{N}\right)^{N-k} \\ &= \frac{\lambda^k}{k!} \cdot \underbrace{\frac{N(N-1)\dots(N-k+1)}{N^k}}_{\rightarrow 1} \underbrace{\left(1 - \frac{\lambda}{N}\right)^{N-k}}_{\rightarrow e^{-\lambda}} \\ &\rightarrow e^{-\lambda} \frac{\lambda^k}{k!} \end{aligned}$$

as $N \rightarrow \infty$. Hence we have

$$\begin{aligned} \Omega &= \{0, 1, 2, \dots\} \\ p_k &= e^{-\lambda} \frac{\lambda^k}{k!} \end{aligned}$$

and we denote this $\text{Poi}(\lambda)$.

§4.2 Random variables

Definition 4.2. Given a probability space $(\Omega, \mathcal{F}, \mathbb{P})$, a *random variable* is a function

$$X : \Omega \rightarrow \mathbb{R}$$

satisfying

$$\{\omega \in \Omega : X(\omega) \leq x\} \in \mathcal{F}$$

for any x . We have the shorthand notation $\{X \leq x\}$ to refer to $\{\omega \in \Omega : X(\omega) \leq x\}$, and $\{X \in A\} = \{\omega \in \Omega : X(\omega) \in A\}$.

Definition 4.3. Let $A \in \mathcal{F}$. Define the *indicator* of A

$$1_A(\omega) = 1(\omega \in A) = \begin{cases} 1 & \omega \in A \\ 0 & \omega \notin A \end{cases}$$

Since $A \in \mathcal{F}$, 1_A is a random variable.

Definition 4.4. Suppose X is a random variable. The *probability distribution function* of X is given by

$$\begin{aligned} F_X(x) &: \mathbb{R} \rightarrow [0, 1] \\ x &\mapsto \mathbb{P}(\{\omega \in \Omega : X(\omega) \leq x\}) = \mathbb{P}(X \leq x) \end{aligned}$$

Definition 4.5. $X = (X_1, \dots, X_n)$ is called a *random variable in $\mathbb{R}^n$* if

$$(X_1, \dots, X_n) : \Omega \rightarrow \mathbb{R}^n$$

is a function such that, for any $x_1, \dots, x_n$,

$$\{\omega : X_1(\omega) \leq x_1, \dots, X_n(\omega) \leq x_n\} \in \mathcal{F}$$

which we denote with the shorthand $\{X_1 \leq x_1, \dots, X_n \leq x_n\}$.

Note that $(X_1, \dots, X_n)$ is a random variable if and only if all of $X_1, \dots, X_n$ are.

Definition 4.6. A random variable X is called *discrete* if it takes values in some countable set S .

Definition 4.7. For a discrete random variable, we are able to write

$$p_x = \mathbb{P}(X = x) = \mathbb{P}(\omega : X(\omega) = x)$$

Then, $(p_x)_{x \in S}$ is called the *probability mass function* of X .

Now we are able to say things like

$$\text{for any } A \subseteq S, \mathbb{P}(X \in A) = \sum_{x \in A} p_x$$

If p_x follows a given distribution, then we say X is a random variable with that distribution or a random variable following that distribution.

Definition 4.8. Let $X_1, \dots, X_n$ be discrete random variables with values in $S_1, \dots, S_n$. They are called *independent* if

$$\mathbb{P}(X_1 = x_1, \dots, X_n = x_n) = \mathbb{P}(X_1 = x_1) \dots \mathbb{P}(X_n = x_n)$$

for all $x_1 \in S_1, \dots, x_n \in S_n$.

Example 4.9

Toss a p -coin N times independently. Then

$$\Omega = \{0, 1\}^N$$

and every $\omega \in \Omega$ is of the form $(\omega_1, \dots, \omega_N)$, where ω_i is the i^{th} outcome. Hence

$$p_\omega = \mathbb{P}(\{\omega\}) = \prod_{k=1}^N p^{\omega_k} (1-p)^{1-\omega_k}$$

Let us also define random variables X_k for each k , satisfying $X_k(\omega) = \omega_k$. Clearly X_k is a Bernoulli r.v. of parameter p . We claim $X_1, \dots, X_N$ are independent: we have

$$\begin{aligned} P(X_1 = x_1, \dots, X_N = x_N) &= \prod_{k=1}^N p^{x_k} (1-p)^{1-x_k} \\ &= \prod_{k=1}^N \mathbb{P}(X_k = x_k) \end{aligned}$$